

Final Report: Analysis of Wind Streaks in SAR Imagery

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1. Project Overview

The goal of this project was to analyze Synthetic Aperture Radar (SAR) images captured by the Sentinel-1 satellite, specifically focusing on the wavelengths and directions of ocean waves associated with wind streaks. These wind streaks are observable on the ocean's surface as wind-driven features that have wavelengths ranging from hundreds of meters to a few kilometers. Using SAR images captured in 25 km by 25 km vignettes with a spatial resolution of approximately 5 m x 5 m, the analysis aimed to calculate both high-frequency and low-frequency wavelengths of the wind streaks, which can provide insights into oceanographic and atmospheric phenomena.

2. Data and Methodology

2.1. Sentinel-1 Mission and SAR Imaging

The Sentinel-1 mission, part of the Copernicus initiative, consists of two sun-synchronous satellites that provide continuous radar imagery for environmental monitoring. In this project, Synthetic Aperture Radar (SAR) images captured by Sentinel-1 were utilized. SAR is an active remote sensing technique where an instrument emits a pulse of energy and records the amount of energy reflected back after interacting with the Earth's surface. These images capture the interaction of radar waves with the ocean surface, enabling the identification of wind streaks.

2.2. Data Preprocessing

The analysis began with preprocessing the SAR images, which involved the following steps:

- **Loading the Image into MATLAB:** The SAR image was imported into MATLAB for further processing.
- **Metadata Extraction:** The dimensions of the image were extracted to calculate the spatial resolution and prepare for subsequent analysis.

- **Grayscale Transformation:** The image was converted to grayscale to simplify the analysis, as grayscale images are typically more suitable for frequency-domain processing.
- **Image Cropping:** The image was cropped to even dimensions, ensuring easier manipulation during analysis, particularly for the Fourier Transform step.

2.3. Fourier Transform

To extract the frequency and direction of the waves, a 2D Fast Fourier Transform (FFT) was applied to the preprocessed SAR images. The FFT decomposes the image into its spatial frequency components, making it possible to identify dominant frequencies and their corresponding directions.

The 2D Fourier Transform can be calculated using the following equation:

$$F(u, v) = \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} f(x, y) \cdot e^{-j2\pi(\frac{ux}{M} + \frac{vy}{N})}$$

The Fourier Transform was then shifted such that the DC component was moved to the center of the spectrum using fftshift:

$$\text{DC Component Location} = \left(\frac{M}{2}, \frac{N}{2} \right)$$

The Magnitude Spectrum was adjusted using a logarithmic transformation for better differentiation between peaks and visualization:

$$\text{Magnitude Spectrum} = c \cdot \log(|\text{Shifted Spectrum}| + 1)$$

2.4. Manual Peak Detection Algorithm

The Manual Peak Frequency Search algorithm can then be applied identify the frequency peaks vis a vis user interaction. The user selects a region within the Fourier-transformed image to focus the frequency analysis on. The process follows these steps:

1. **User Range Selection:** The user manually selects a specific range within the Fourier-transformed image that corresponds to the area of interest for frequency analysis.
2. **Search within Selected Range:** The algorithm extracts the magnitude spectrum from the selected region.

3. **Find Maximum Peak:** The algorithm then scans the cropped region of the magnitude spectrum to identify the dominant peak, which represents the primary frequency component.
4. **Frequency and Direction Calculation:** Once the peak is located, its coordinates are used to compute the frequency components in both the x and y directions (fx, fy). From these components, the wavelength and direction of the wave are derived.

2.5. Automatic Peak Detection Algorithm

The Automatic Peak Frequency Search algorithm can be applied as well. It utilizes a series of filters and processing steps to identify dominant frequency components in the image. The process is outlined below:

1. **Low-Pass Gaussian Filter Creation:** A low-pass (LP) Gaussian filter is designed in the frequency domain. The filter is defined as follows:

$$H_{LP}(u, v) = \exp \left(-\frac{(u - \frac{N}{2})^2 + (v - \frac{M}{2})^2}{2\sigma^2} \right)$$

where U and v are the frequency indices, N and M represent the image dimensions, and σ controls the filter width.

2. **High-Pass Image Calculation:** The high-pass (HP) image is derived by subtracting the LP filter from the original frequency-domain image. This process isolates the high-frequency components, while the low-frequency components are removed:

$$H_{HP}(u, v) = 1 - H_{LP}(u, v)$$

3. **Filtering the Image:** The original image is processed through both the low-pass and high-pass filters to generate two separate images:
 - **Low Frequency Image:** Captures the low-frequency components.
 - **High Frequency Image:** Captures the high-frequency components.
4. **Low-Pass Filter for DC Component Suppression:** A very low-pass filter/mask is applied to the low-frequency image to reduce leakage from the DC peak. This step helps ensure that low-frequency peak detection is not distorted by the influence of the DC peak.
5. **Peak Detection and Averaging:** The low frequency and high frequency images are divided into a predefined number of windows. The algorithm identifies the maximum values within each window of the filtered images.

5. **Frequency and Direction Calculation:** The coordinates are used to compute the frequency components in both the x and y directions (f_x , f_y) for each window. The average frequency components are calculated and from them the average wavelength and direction of the wave are derived.

2.6. Wavelength and Direction Calculation

The **wavelength** and **direction** were calculated based on the Fourier Transform peak coordinates in the x and y directions.

The frequency components in the x and y directions were calculated from the peak coordinates using the following equation:

$$f_x = \frac{u}{M \cdot \Delta x} \quad f_y = \frac{v}{N \cdot \Delta y}$$

where U, V are the frequency indexes, M, N is the total number of points (pixels), and Δx , Δy is the spatial resolution (distance between pixels).

Since the fft was shifted, the u and v indexes are adjusted as follows:

$$u = coordinate_x - \frac{M}{2}, \quad v = coordinate_y - \frac{N}{2}$$

Finally, the wavelength magnitude was calculated as:

$$\lambda = \frac{1}{f} = \frac{1}{\sqrt{f_x^2 + f_y^2}}$$

Where λ is the wavelength in the same units as the spatial dimensions of the image (e.g., meters if the image pixel spacing is in meters) and f_x and f_y are the spatial frequencies in the x and y directions.

The wave direction was calculated as:

$$\theta = \tan^{-1} \left(\frac{f_y}{f_x} \right)$$

Where θ is the direction of the wave relative to the x-axis, measured in radians or degrees and f_x and f_y are the spatial frequencies in the x and y directions.

2.7. Challenges in Analysis and Solutions

- **Lower Resolution in PNG Images:** The lower resolution of PNG images posed a challenge in identifying precise frequency peaks. The limited detail in the images made it difficult to accurately detect and analyze fine frequency components.
- **Spatial Frequency Variability:** The frequency components across the image exhibited significant variability, creating a "cloud" of peaks rather than distinct, isolated frequency peaks.
 - **Solution:** To address this, a windowing technique was applied. This method averages the peaks across different image regions.
- **Low-Frequency Challenges:** Peaks near the DC component were often hard to distinguish from DC leakage, which introduced distortions and affected the accuracy of low-frequency peak detection.
 - **Solution:** A zeroing mask with a small radius was applied around the DC component to suppress leakage, ensuring that the DC component did not interfere with the identification of valid frequency peaks. This method helped, but more advanced filtering techniques could provide better results.
- **Inverse Relationship Between Frequency and Wavelength:** Small shifts in the low-frequency values resulted in substantial changes to the calculated wavelengths, highlighting the sensitivity of wavelength estimation to frequency variations.
- **Parameter Tuning:** Identifying the optimal window lengths and Gaussian filter width (σ) required a process of experimentation and adjustment. Iterative fine-tuning was necessary to ensure the accuracy of the results.
 - **Solution:** Window sizes and filter parameters were adjusted iteratively, and the filtered images were checked to verify the effective separation of high- and low-frequency components.
 - Despite these efforts, small changes in parameters such as window size, filter width, and the radius of the zero mask around the DC component still significantly impacted the identified peak frequencies, causing variations in the results.

3. Results

3.1. High-Frequency Wavelengths

- **Range** ~ 100 m to 300 m
- These wavelengths are typical of open ocean waves and were observed in both wind streaks and in areas without visible streaks.
- The results aligned with expected ocean wave patterns.

3.2. Low-Frequency Wavelengths

- **Range** ~ 900 m to 2.6 km

- These values align with theoretical predictions for wind streaks, with longer wavelengths indicating higher wind speeds.
- Resolution limitations and DC leakage affected the accuracy of identifying **precise** magnitudes for low-frequency wavelengths

4. GUI Development

A Graphical User Interface (GUI) was developed to streamline the analysis process. This interface allows users to:

- **Upload and Display SAR Images:** Users can easily upload and visualize the SAR images for analysis.
- **Perform Fourier Analysis:** The GUI facilitates Fourier analysis, allowing users to view the resulting magnitude spectrum.
- **Visualize Peak Magnitudes:** Users can view the peak magnitudes as points on the magnitude spectrum
- **View Wavelengths and Directions:** The detected frequencies' wavelengths, intensities, and directions can be visualized directly on the interface as arrows, where the length of each arrow corresponds to the peak intensity and the direction of propagation is indicated. This can be overlaid on the magnitude spectrum, the original image or the filtered images.
- **Use Binary Thresholding:** A binary thresholding tool is available to aid in identifying peaks within the Fourier transform.
- **View Filtered Images:** After performing automatic peak detection, users can view both low- and high-frequency filtered images.
- **Edit Image Resolution:** The GUI allows users to adjust image resolution and view the image dimensions and pixel resolution for better control and analysis.

5. Conclusion

This project utilized Synthetic Aperture Radar (SAR) imagery from the Sentinel-1 mission to analyze wind streaks over the ocean, focusing on calculating the associated high- and low-frequency wavelengths and their directional orientations. Through the application of Fourier analysis, the dominant frequencies were identified.

The development of a user-friendly graphical interface (GUI) facilitated the analysis by enabling users to easily upload images, perform Fourier transforms, and visualize the results. Despite challenges posed by the lower resolution of PNG images and complexities associated with frequency variability and DC leakage, the project achieved its primary goal of detecting and analyzing the wavelengths of ocean waves influenced by wind streaks.

The results showed that low-frequency wavelengths ranged from 900 meters to 2.6 kilometers, consistent with theoretical predictions for wind streaks, while high-frequency wavelengths aligned with typical ocean wave patterns (around 100-300 meters).

Challenges in low-frequency analysis were mitigated by using filtering and windowing techniques. These challenges illustrate the complexities of frequency analysis in SAR images, emphasizing the need for careful parameter tuning and handling of resolution limitations and DC leakage. More advanced low frequency filtering techniques and images with better resolution would help improve the accuracy and stability of wavelength calculations.

Overall, the project demonstrated the relevancy of SAR imagery and Fourier analysis in oceanographic image analysis.

6. References

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